Low-Light Video Enhancement via Negative Image Dehazing
透過負影像去霧技術改善低光影片品質

Arthur-king.Kao (高塏翔)

Department of Mathematics, National Central University
Jhongli District, Taoyuan City 320317, Taiwan

August 5, 2026

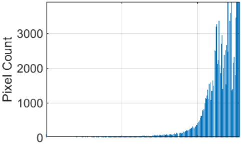
Motivation

In daily life, we often encounter underexposed or low-light images. This led to the hypothesis that by applying an inverse transformation (negative) to a Low-Light image, it would statistically resemble a hazy image. Consequently, we can leverage established image dehazing techniques to restore visibility and details.

Histogram

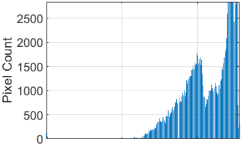
Low-Light Image

Negative Image



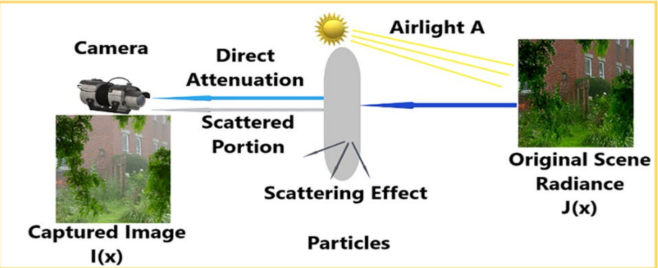
Negative Image of hist

Hazed Image



Hazed Image pf hist

Atmospheric Scattering Model(ASM)



The **Atmospheric Scattering Model** is a physical model that describes how light attenuates and scatters in environments filled with particles.

$$I(x) = J(x)t(x) + A(1 - t(x))$$

- $I(x)$: Captured low-light image
- $J(x)$: Recovered scene radiance
- $t(x)$: Transmission map
- A: Global atmospheric light

*Mapped to negative domain for dehazing.

Dark Channel Prior (DCP)

The Dark Channel Prior is based on the observation that in most non-haze outdoor images, at least one color channel has some pixels whose intensity is very low and close to zero:

$$\mathcal{J}_{dark}(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{R,G,B\}} \mathcal{J}^c(y) \right) \approx 0$$

Why is it close to zero?

- **Shadows:** Natural shadows in images (e.g., crevices, occlusion).
- **Colorful Objects:** Vivid colors (e.g., a red apple) have very low values in other channels (blue/green).
- **Dark Objects:** Naturally dark surfaces like tree trunks or dark soil.

Estimation of $t(x)$ and A

1. Transmission Map $t(x)$

Derived by assuming constant transmission $\tilde{t}(x)$ in local patch $\Omega(x)$ and applying DCP ($J^{dark} \approx 0$):

$$\tilde{t}(x) = 1 - \omega \min_{y \in \Omega(x)} \left(\min_{c \in \{R,G,B\}} \frac{I^c(y)}{A^c} \right)$$

- Calculated patch by patch.
- In this project, $\omega = 0.95$ to retain a small amount of haze for natural depth.

2. Atmospheric Light A

Estimated using the Dark Channel to avoid interference from bright white objects:

1. Pick the **top 0.1% brightest** pixels in the dark channel image J^{dark}.
2. These represent the most haze-opaque regions.
3. Identify corresponding pixels in input image I.
4. Select the **average intensity** among them as A.

Guided Filter

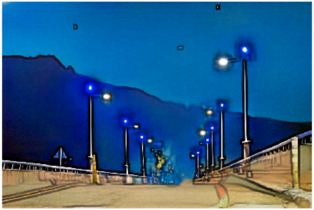


Figure: Blocky Artifacts

The **Guided Filter** is an edge-preserving smoothing filter. It uses a "Guidance Image" to determine the filtering output, allowing the output image to inherit the structural and edge information of the guidance image.

Why use Guided Filter?

The initial transmission map $t(x)$ is calculated patch by patch, which often leads to blocky artifacts near sharp edges.

Restoring J(x)

After estimating the global atmospheric light A and refining the transmission map

$$t(x) = \tilde{t}(x) \rightarrow q$$

via the Guided Filter, we can finally recover the clear scene radiance $J(x)$ by inverting the Atmospheric Scattering Model:

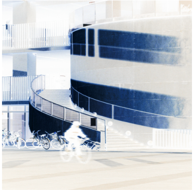
$$J(x) = \frac{I(x) - A}{\max(t(x), t_0)} + A$$

Why use max and What is t_o?

We set $t_0 = 0.1$ to prevent $t(x)$ from being too close to 0, which would make the division value too large.

Result


Low-Light Image


Negative Image


Dark Channel


Resulting Image


+Guided Filter


Video Result.

Reference

1. X. Dong, et al., " Fast efficient algorithm for enhancement of low lighting video ", *2011 IEEE International Conference on Multimedia and Expo, Barcelona*, 2011, pp. 1-6
2. K. He, J. Sun, and X. Tang, " Single image haze removal using dark channel prior ", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33 (2011), pp. 2341-2353
3. S.-Y. Yang, " Single image dehazing, Lecture Slides ", *Lecture Slides*, 2025